Hussain Singapuri

AI ENGINEER (PYTHON, LLM & AGENTS)

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PROFESSIONAL SUMMARY

AI Engineer with 3 years of professional full-stack experience at a US-based medical-device company, building **AI agents**, a CRM and web apps. Builds **LLM** applications in **Python** with **function calling**, multi-agent orchestration, **RAG** (hybrid BM25 + **pgvector** vector database) and eval harnesses, exposed through **REST APIs** (FastAPI) and secured integrations. Ships with **Docker** and GitHub Actions CI. Uses **Claude Code** daily. Claude Certified Developer. Bachelor's in Computer Science (AI & Machine Learning).

CORE COMPETENCIES

LLM Applications & AI Agents | Python Backend (FastAPI, asyncio) | RAG & Vector Databases (pgvector) | Function Calling & API Integrations | AI Orchestration (multi-agent DAG) | Prompt Engineering & Evals | Docker & CI/CD | Security & Secure Integrations

WORK EXPERIENCE

NeuroNetworks AI Ed

2026 - Present

CTO & Principal Course Architect

Pune, India

- Built and run the website neuronetwrks.com with a Python static-site generator deployed on Vercel, including programmatic SEO
- Lead technology for an Indian AI and tech education brand (Neuronetworks AI Ed Private Limited) offering programmes for students after Class 10
- Architected the curriculum: a 1-year AI Foundation course and five 3-year specialisations (AI/ML Engineering, Software Development, Data Science & AI, Product & UX / AI Design, Cybersecurity)

US-based medical-device company

Mar 2023 - Sep 2026

Full Stack Developer

Remote

- Built **AI agents** for internal use
- Built and maintained full-stack web applications for the company
- Built several apps, including a **CRM**
- Performed penetration / security testing on 5+ of the company's mobile apps, all live on the Google Play Store

PROJECTS

multi-agent-research-crew

Open Source

LLM orchestration in Python: planner, parallel researchers, writer and verifier agents on a custom DAG orchestrator with retries, timeouts, token budgets and claim-level citation checking.

Python, asyncio, Claude

site-to-agent

Open Source

RAG pipeline: crawler, hybrid BM25 + vector retrieval (pgvector) with RRF, cited answers, SSRF-hardened fetching and an eval harness.

Next.js, TypeScript, pgvector

nanoserve

Open Source

LLM inference engine in Python/PyTorch: paged KV cache, continuous batching, speculative decoding, OpenAI-compatible streaming API (FastAPI); 3.1x throughput over sequential decoding.

Python, PyTorch, FastAPI

ai-lead-receptionist Open Source

AI agent (Claude) integrated with Twilio SMS and web chat: qualifies leads and books jobs through API integrations, with an owner dashboard.

Next.js, TypeScript, Twilio, Claude, Supabase

lead-intel-engine Open Source

Python/FastAPI service that discovers, enriches and scores local-business leads with a robots-aware, rate-limited crawler and transparent scoring.

Python, FastAPI, SQLite

PageForge Portfolio project

Responsible-AI controls: AI drafts only from approved facts, a deterministic validator flags unsupported numbers, a reviewer signs off claim by claim.

Next.js 16, React 19, Tailwind v4, Sanity, Supabase, Gemini, PostHog, HubSpot

More projects

contractor-ai-ops-dashboard, Orbital Surplus, Nightjar & Co., Lumen, Kilnwood Roasters, Cairnmoor, Lowlight Video Club, Safey Inventory, TripSync, personal portfolio

EDUCATION

Bachelor's, Computer Science (AI & Machine Learning) Vishwakarma Institute of Technology, Pune 2022-2025

Pune

CERTIFICATIONS

Claude Certified Developer	Anthropic, Foundations (CCDV-F)
Claude Certified Orchestrator	Anthropic, Foundations (CCAO-F)
Generative AI Leader	Google Cloud
Claude Code in Action	Anthropic
Claude Code 101	Anthropic

SKILLS

AI / LLM: LLM applications, AI agents, tool use / function calling, RAG (hybrid BM25 + vector, RRF), multi-agent orchestration, prompt engineering, evals, prompt-injection defence, Claude API, Gemini API, LLM inference serving, Claude Code and AI coding agents

Languages: Python, TypeScript, JavaScript, SQL, Dart, HTML, CSS

Back end: Python (FastAPI, asyncio), REST APIs, Node.js, webhooks (HMAC signing), auth (scrypt, sessions, throttling), Stripe Checkout and refunds, Resend

Data: PostgreSQL, Neon, Supabase (RLS, Edge Functions), pgvector, PGlite, SQLite, Sanity CMS

DevOps: Docker, GitHub Actions, Vercel

Automation & integrations: workflow automation, CRM integrations (HubSpot), PostHog, Twilio SMS, web crawling / scraping

Security: mobile app penetration testing, CSP and security headers, SSRF hardening, input validation

Front end: React 19, Next.js, TypeScript, JavaScript, Tailwind CSS, shadcn/ui, TanStack Start / Query, GSAP, Lenis, three.js, WebGL, Canvas 2D, responsive and accessible UI, motion and interaction design

Mobile: Flutter, Riverpod, Firebase **Design:** UI/UX design, design systems, technical SEO, JSON-LD structured data, programmatic SEO